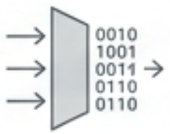
Module 12, Lesson 1

AI 101 for Mechanical Engineers: Represent

Represent the information before choosing a network.

Gemini Notebook



1. Data Representation Modalities



2. Neural Network Architectures



3. Evaluation and Data Splitting

1

Section 1: Data Representation Modalities

Purpose: Define structural formats for engineering data inputs.

Gemini Notebook

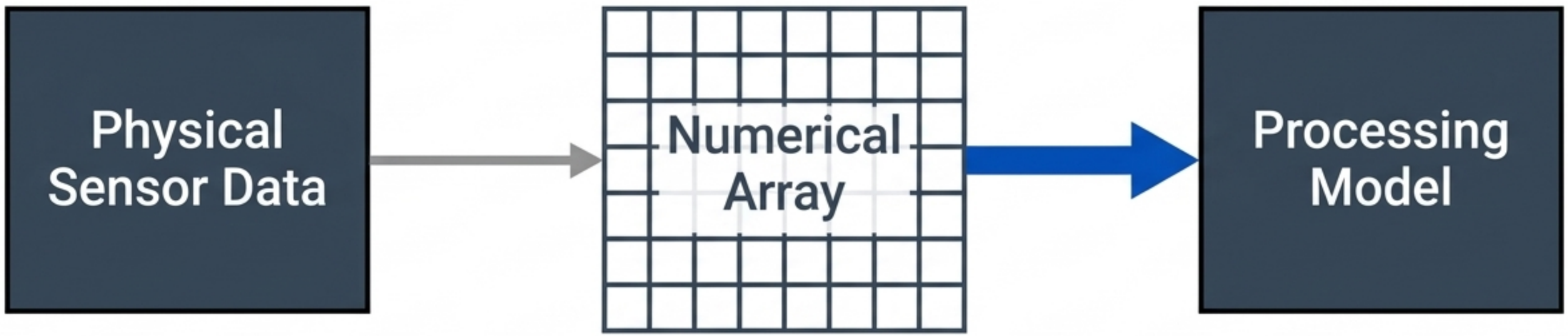
Fundamental Data Representations

Representation	Example	Relevant issue
Feature vector	Explicit operating features	Meaningful inputs
Image	Pixels and channels	Lighting and component identity
Signal	Timed physical samples	Units and sampling rate

- Tabular data utilize explicit feature vectors.
- Images represent spatial values via pixels.
- Signals record sample timing and units.
- Time series index observations by time.

Gemini Notebook

Representation Limits



- Conversion generates pure numerical arrays.
- Process defines mathematical data representation.
- Representation guarantees no engineering comprehension.
- Models require explicit meaning definition.

Gemini Notebook

2

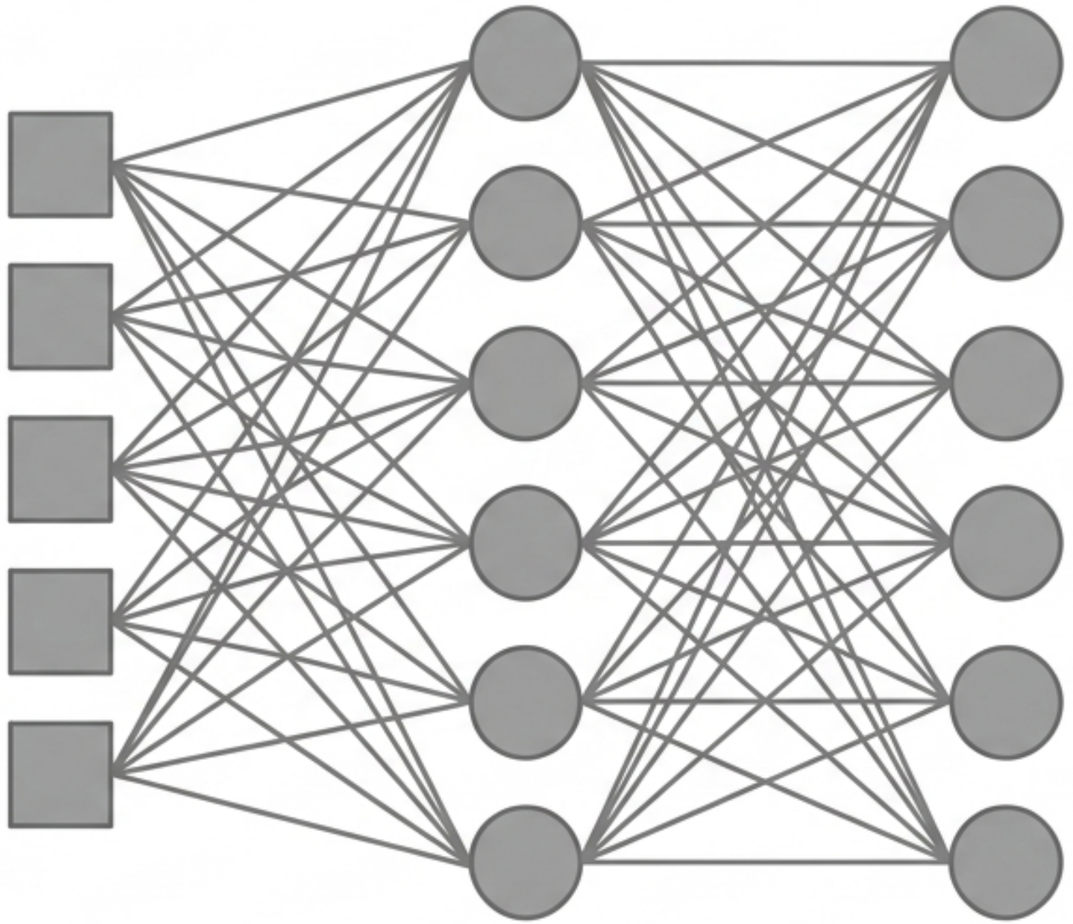
Section 2: Neural Network Architectures

Purpose: Compare MLP and CNN processing mechanisms.

Gemini Notebook

Multilayer Perceptrons (MLP)

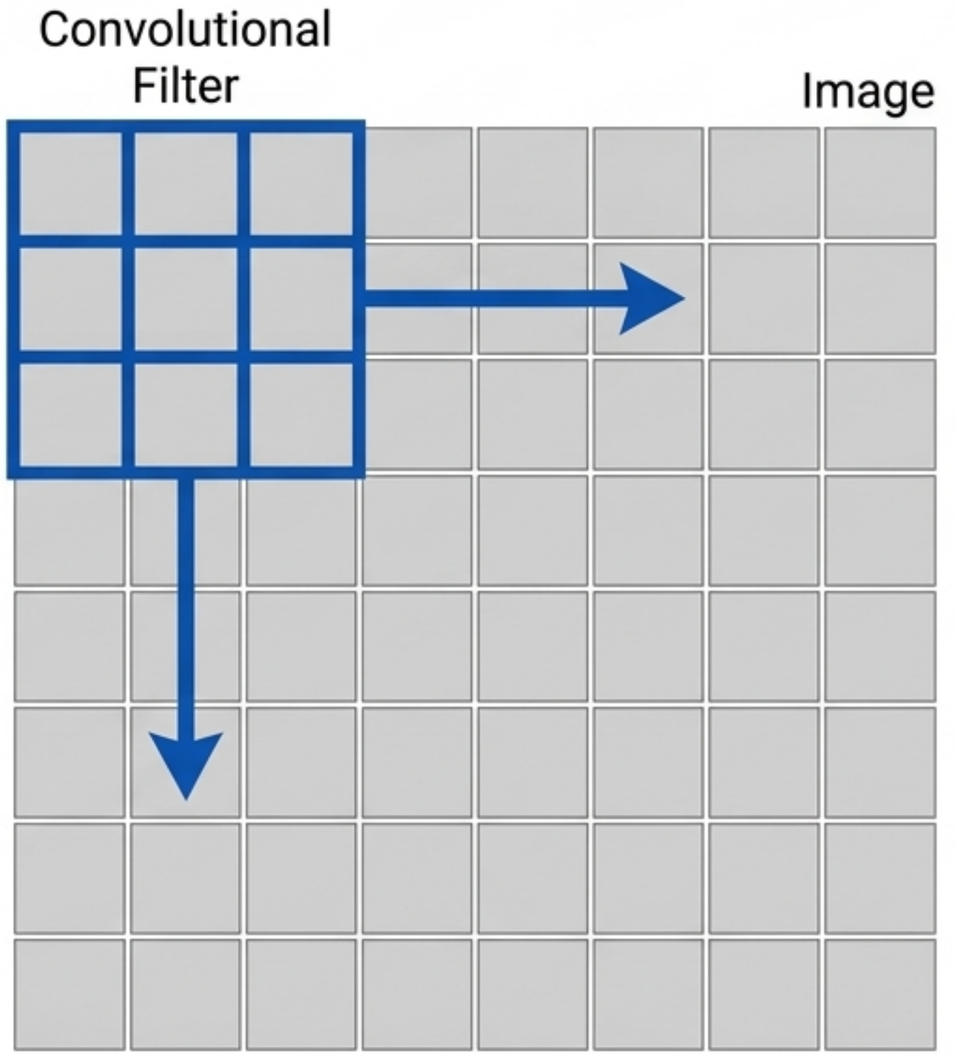
- Utilizes fully connected neural layers.
- Operates on explicit feature vectors.
- Predicts outcomes from standard tabular inputs.



Gemini Notebook

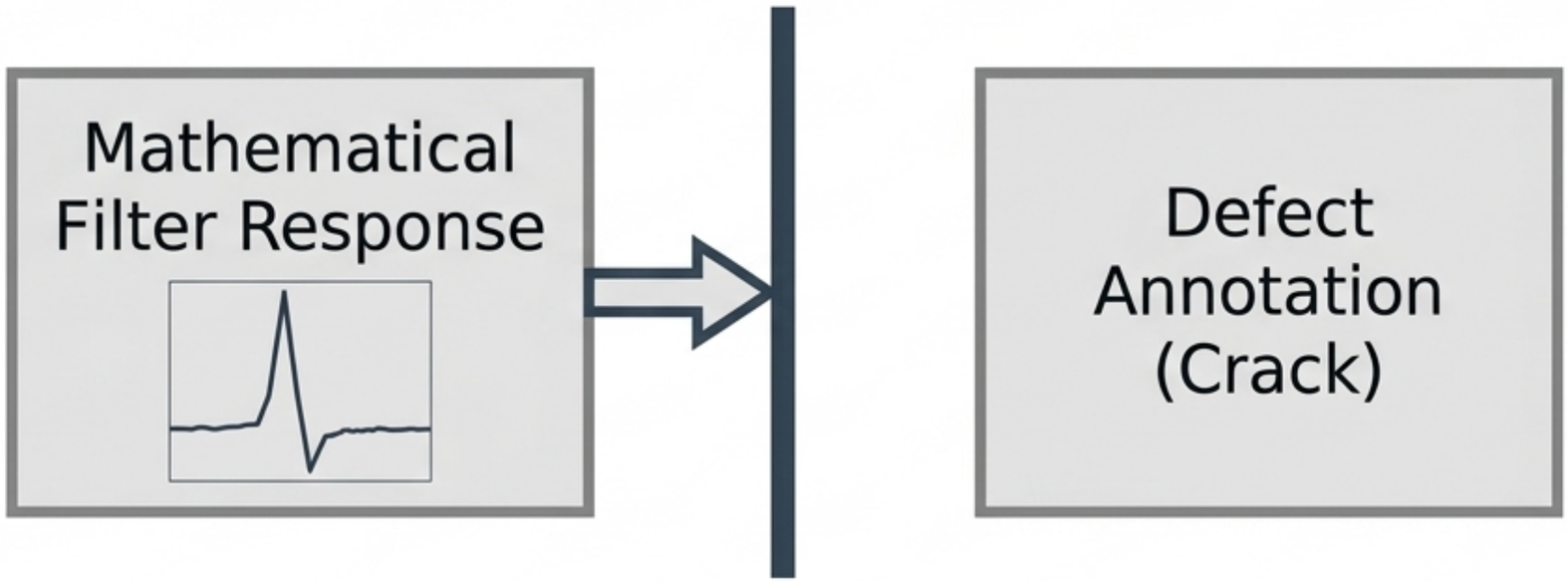
Convolutional Neural Networks (CNN)

- Utilizes shared convolutional filters.
- Computes local spatial patterns.
- Processes images and specific signals.
- Detects edges and surface features.



Gemini Notebook

Evidence and Interpretation



- Filters respond to local geometry.
- Matheinatical responses lack inherent classification.
- Responses require external labelled evidence.
- Causation requires explicit engineering validation.

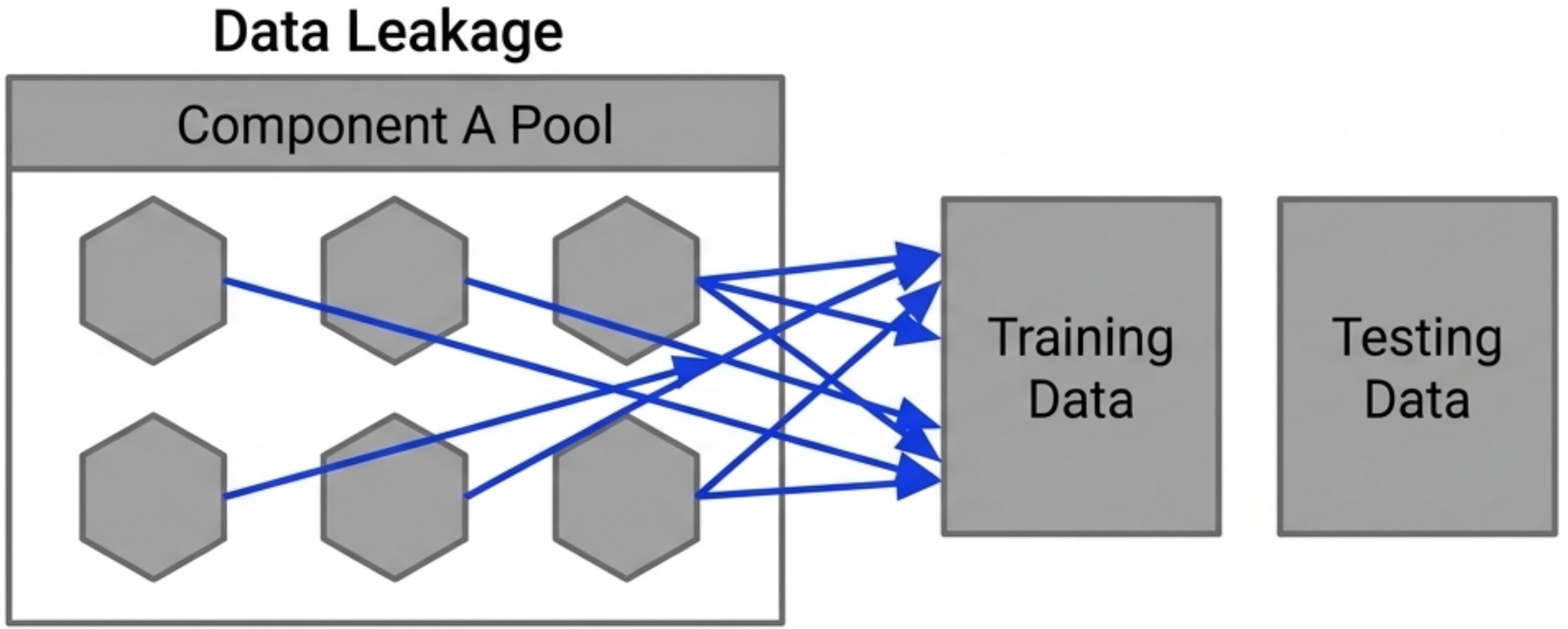
Gemini Notebook

3 Section 3: Evaluation and Data Splitting

Purpose: Establish robust testing protocols for new components.

Gemini Notebook

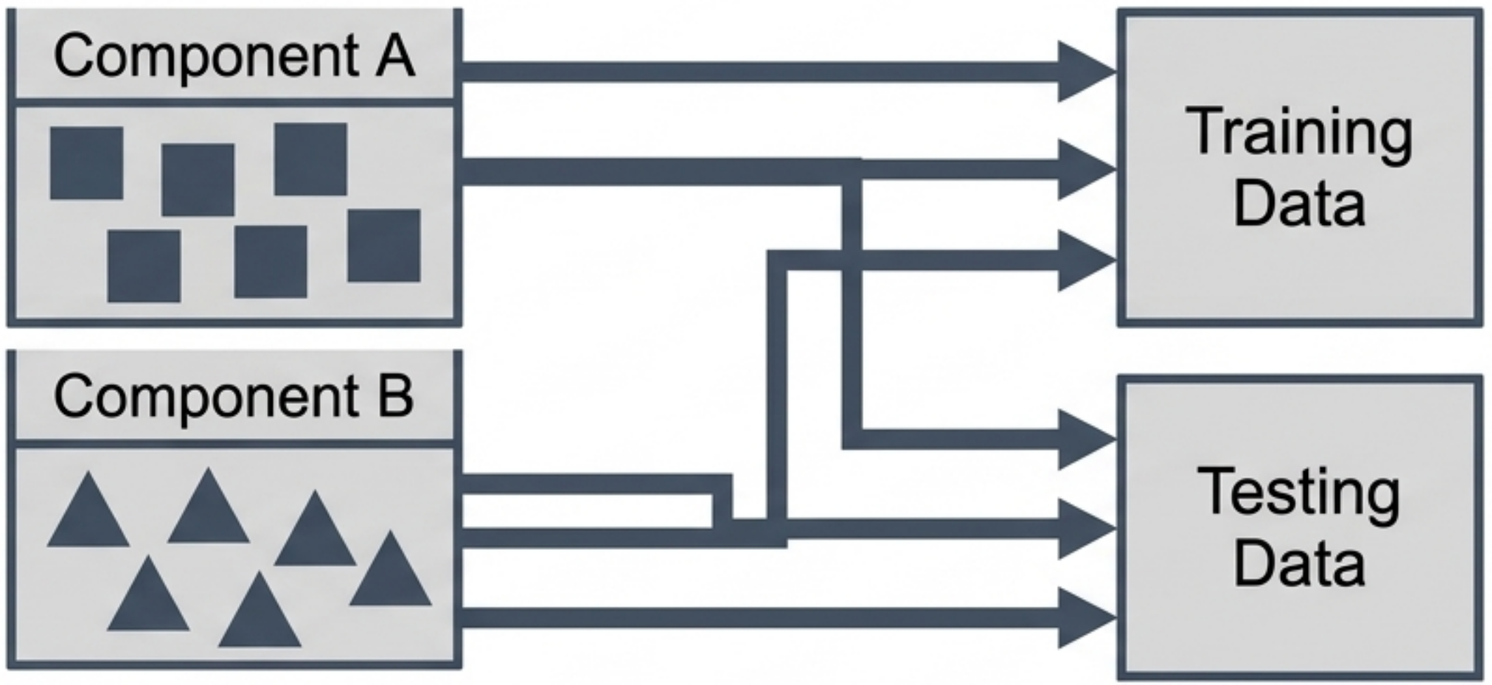
The Image Split Problem



- Random splitting distributes identical component images.
- Shared surface patterns leak across splits.
- Lighting configurations leak across splits.
- Performance metrics become artificially overstated.

Gemini Notebook

Robust Holdout Strategies



- Isolate complete components for testing.
- Isolate unique batches for validation.
- Evaluate specific new-component inspection tasks.
- Account for annotation and camera variations.

Gemini Notebook

Key Points

1. Representation defines data structure, not engineering meaning.

2. Architecture dictates processing mechanism (MLP vs CNN).

3. Component-level holdouts prevent evaluation data leakage.

Gemini Notebook